

RFID HACKING

wait... it's that easy!?

WHO'S THIS?

- Telecoms and networking geek
- Infrastructure and Security
- RFID-curious for ~5 years
- Good tools and visibility made all the difference
- Today's motivation:
 - Surface the simplicity and vulnerability of supposedly-secure systems
 - Give you tools and information to follow your own curiosity
 - Hack in to the office!?
 - Not really, but essentially the same thing.

RFID

- Contactless devices using Radio Frequency (RF) energy to communicate
 - Can mean a lot of things: military IFF, inventory markers, building access keys, etc.
- With some identifying characteristic
 - Usually a bitstream or number
- Radio Frequency, Identification
- Our scope in here: cards and fobs used to access digitally-controlled physical systems

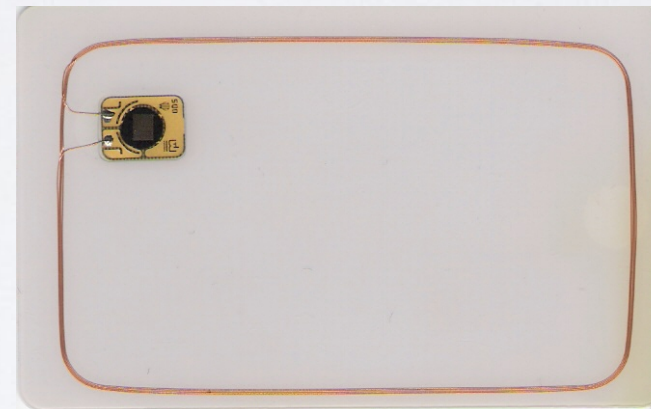
INTERFACE TYPES

- Low Frequency (LF) -- 125-135 kHz (λ in kilometers)



- LC coil; long and thin wire coil

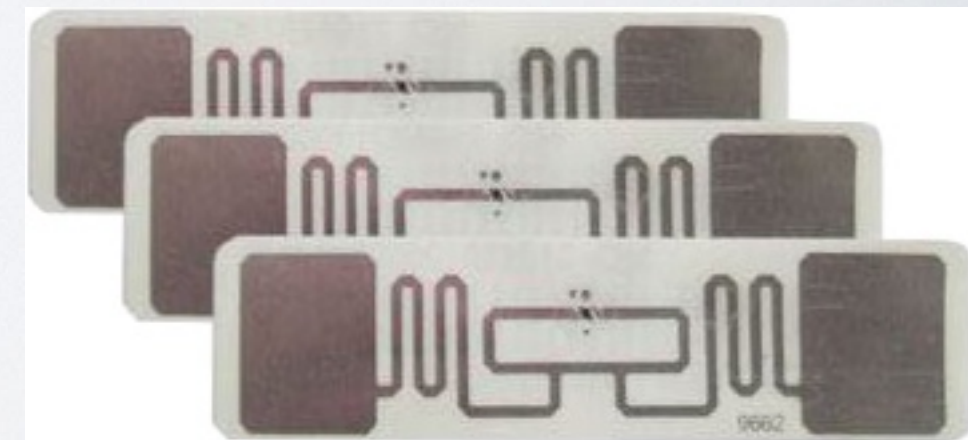
- High Frequency (HF) -- 13.56 MHz (λ in meters)



- LC coil; less-long, thin wire coil

- Ultra High Frequency (UHF) -- 850-950 MHz (λ in centimeters)

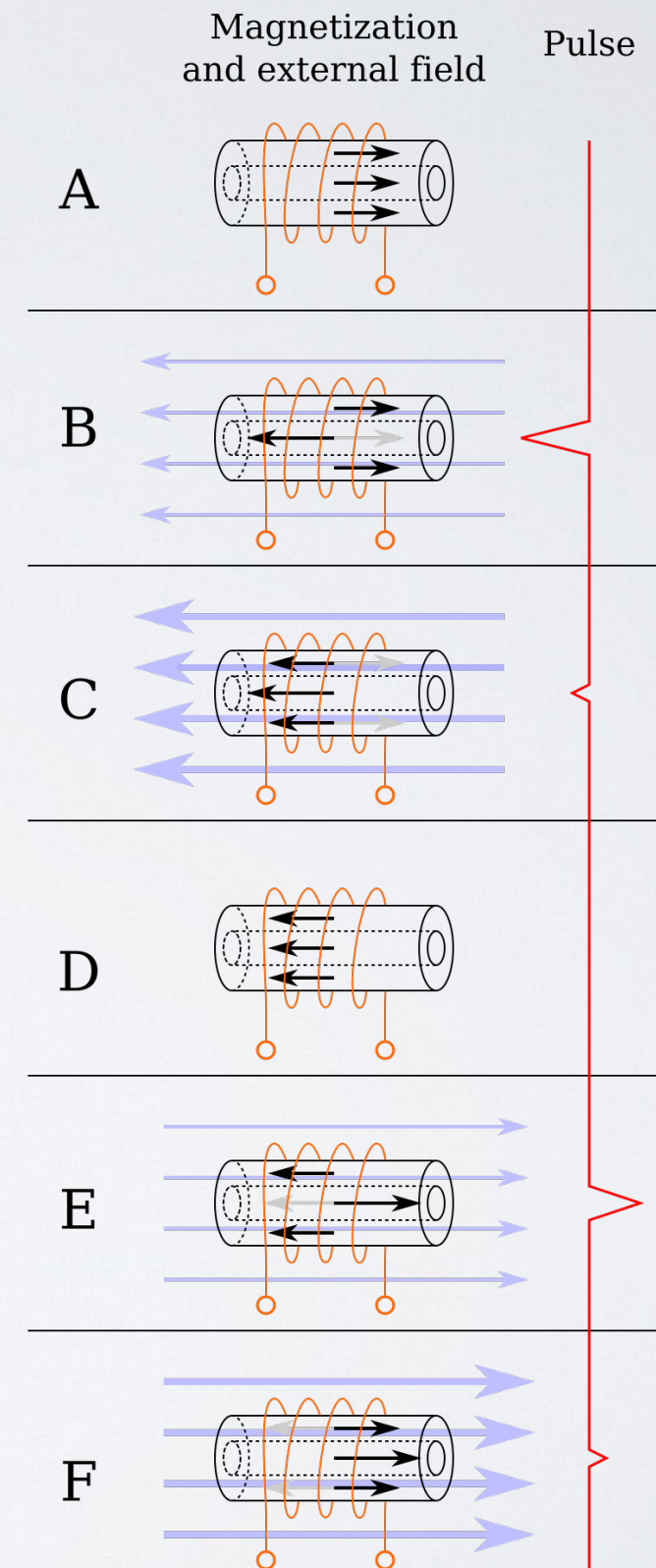
- Dipole Antenna

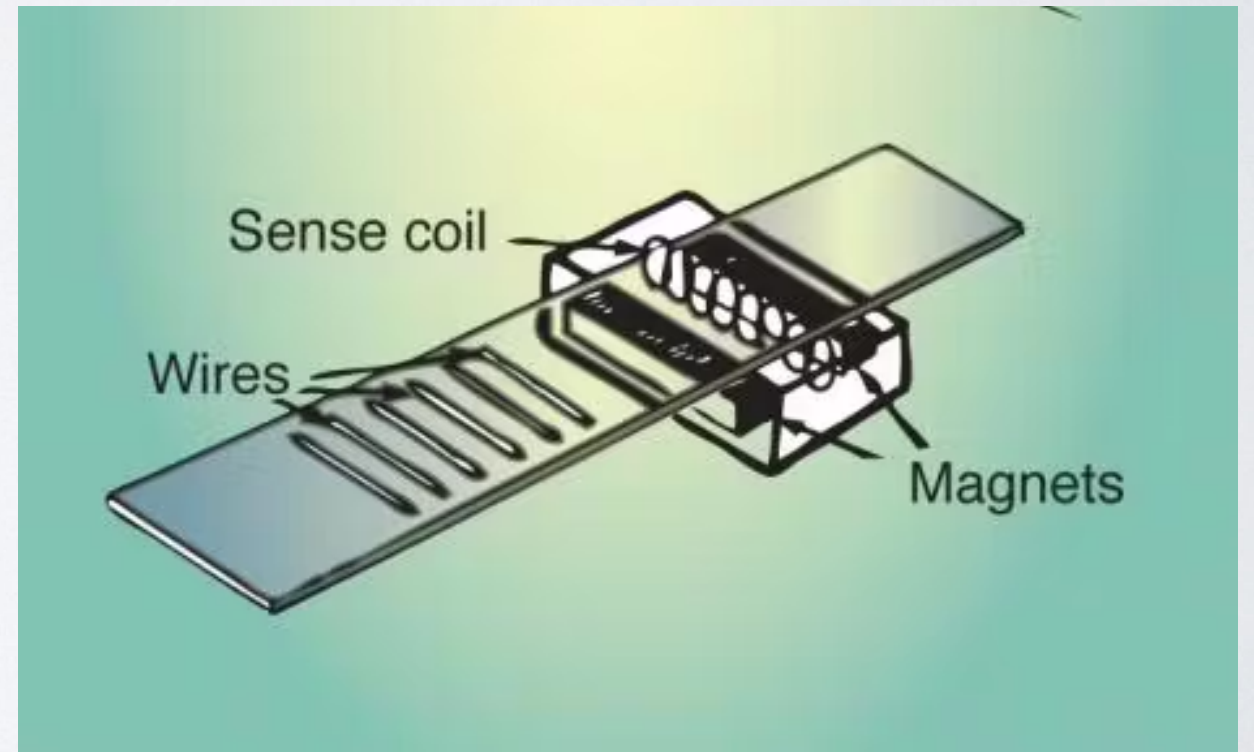


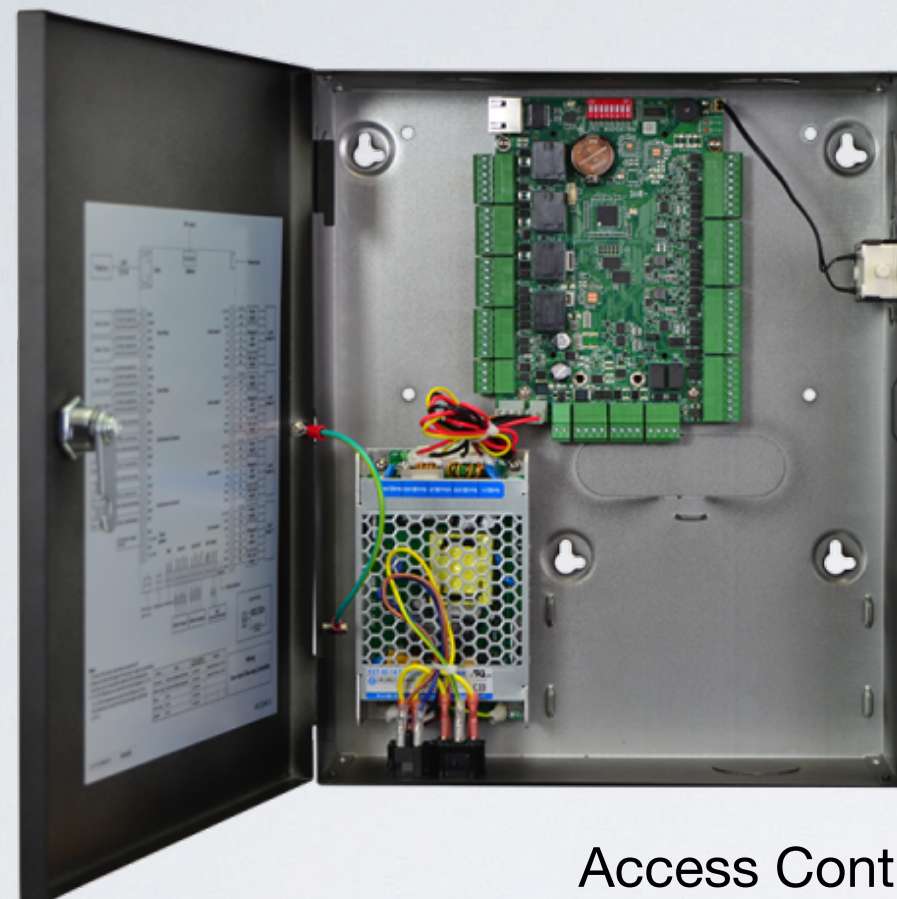
Broadly: when using compact antennas, the higher the frequency, the longer the range, but the costs are also higher

HISTORY: WIEGAND CARDS

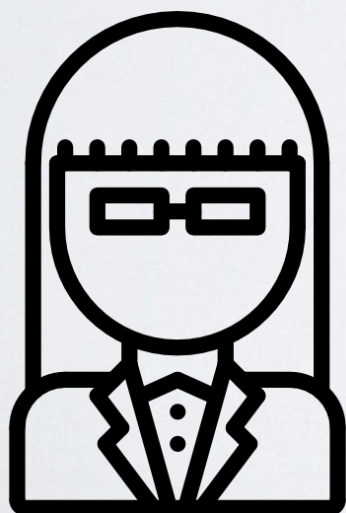
- John R. Wiegand discovers the Wiegand Effect
 - An annealed metal wire with different magnetic coercivities in the core vs. outside.
 - Creates a non-linear “snapping” of magnetic domain alignment in the presence of an external field, quickly reversing.
 - Useful, as this fast reversal creates a nice voltage spike signal in a detection coil.
- Patented 1974



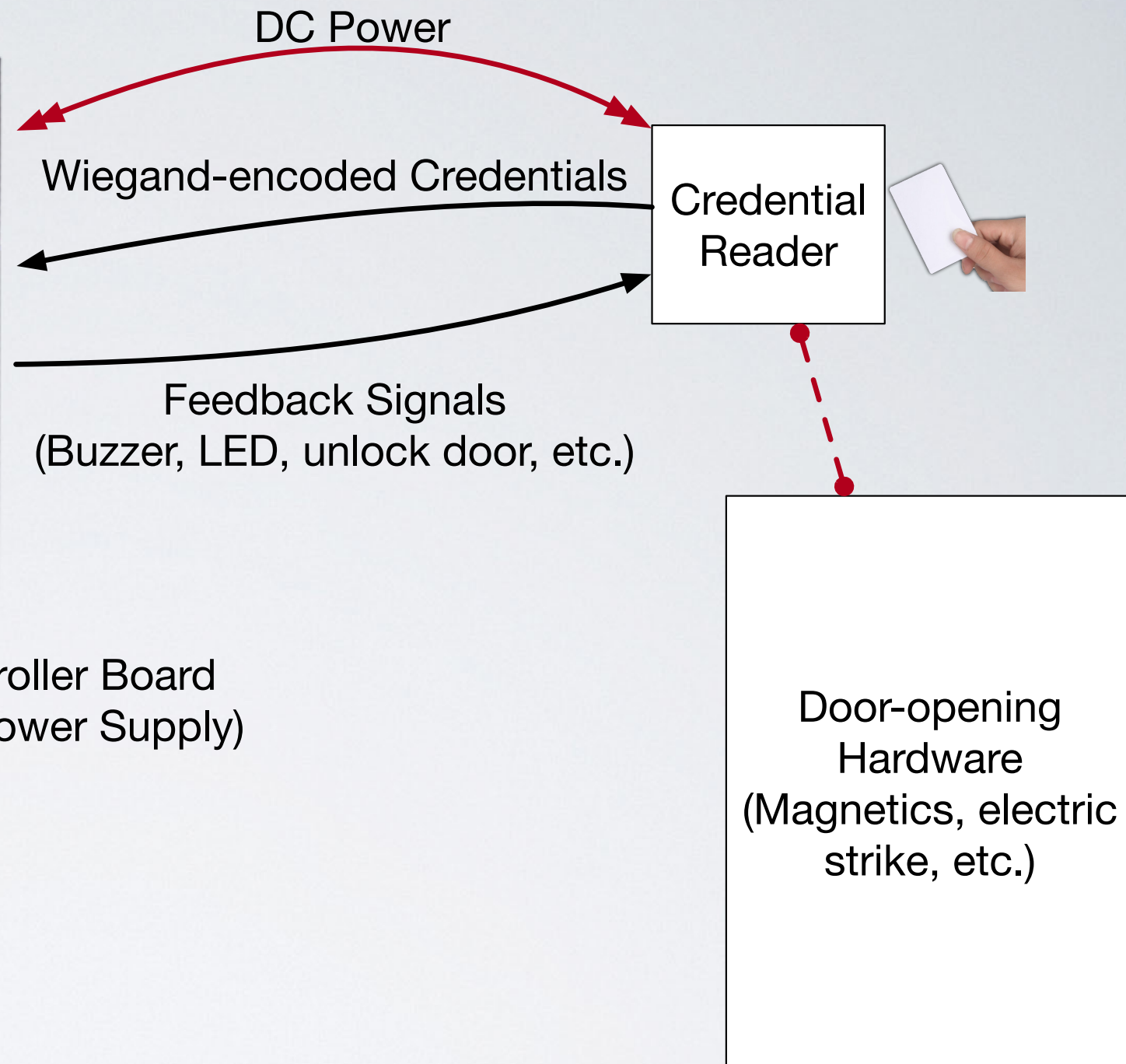


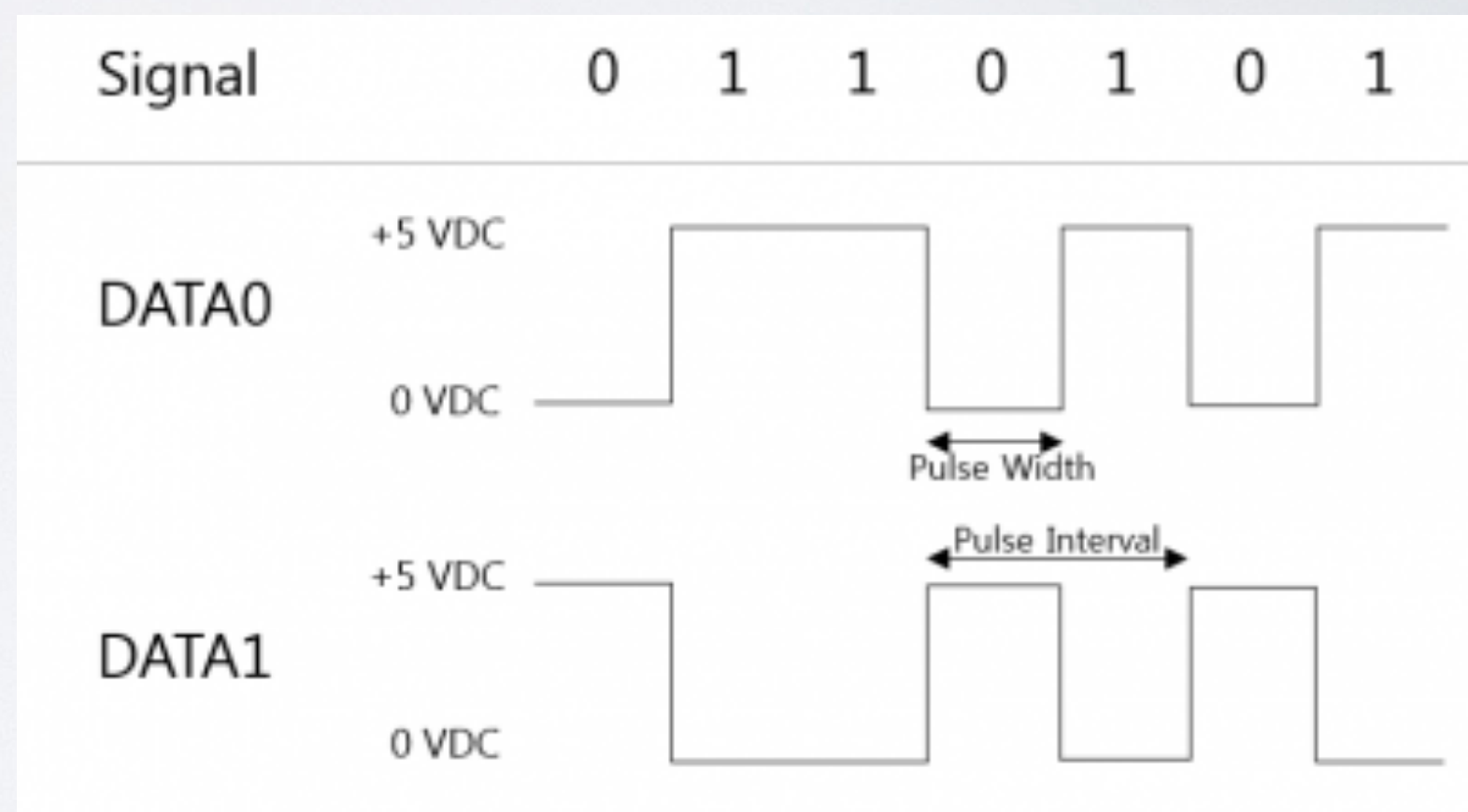
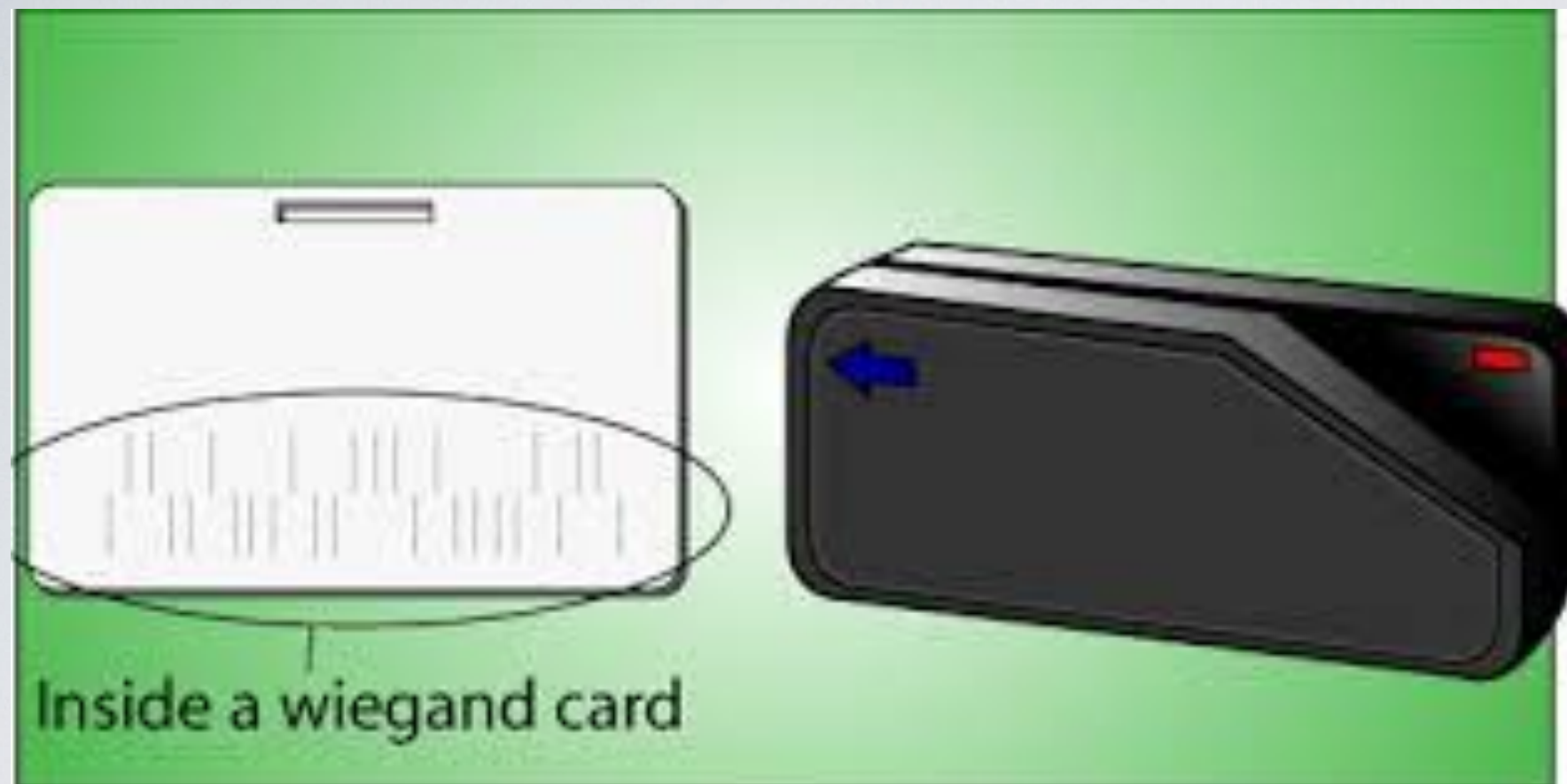


Access Controller Board
(and AC-DC Power Supply)



Administration and Enrollment





8 bits of Facility Code

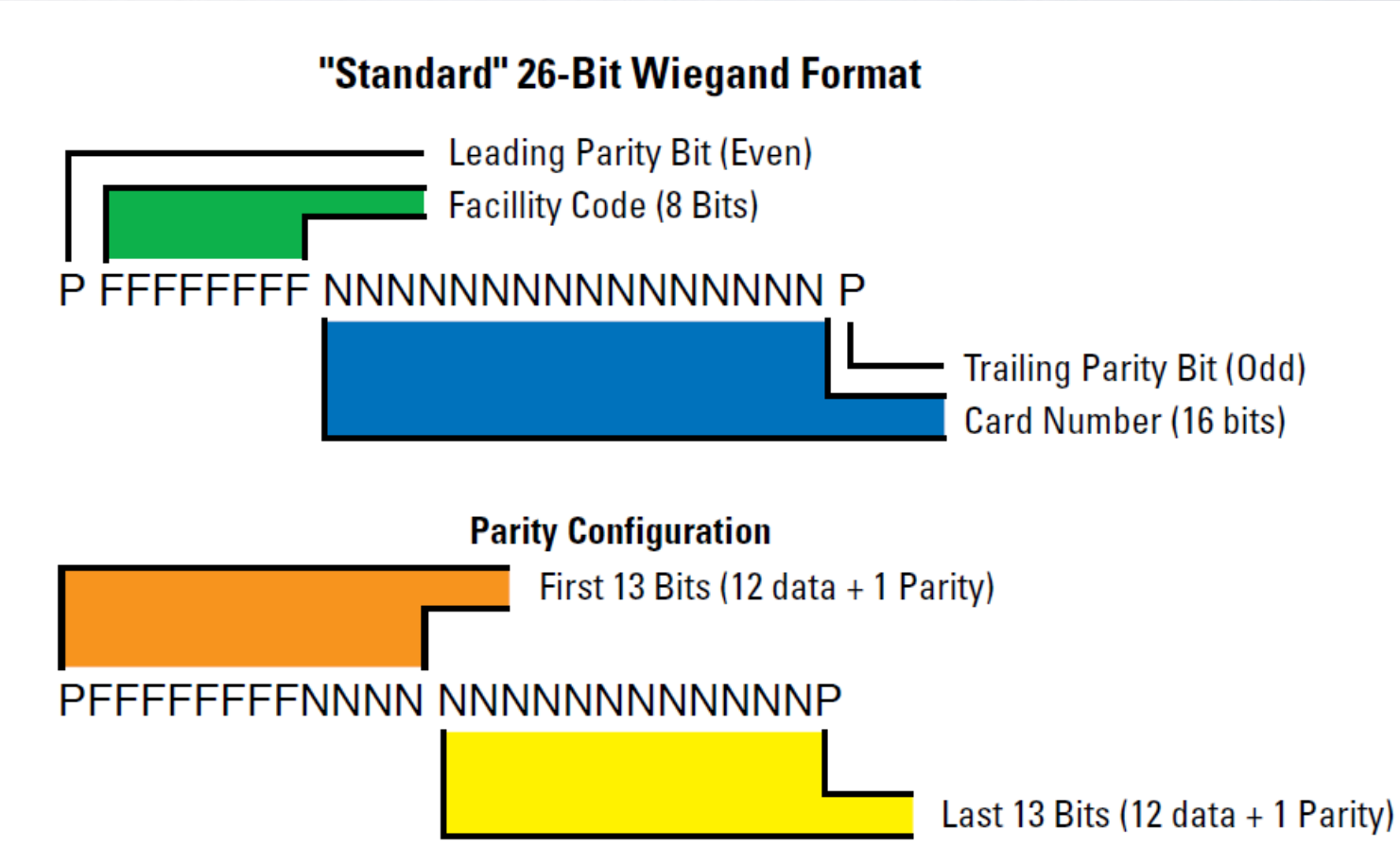
16 bits of Card ID Number

This *looks* like 24 bits of entropy,
but in practice is much less.

Facility Codes are reused

Administrators tend to order
pre-programmed cards in
numeric ranges

26-Bit Wiegand Format	
Bit Number	Purpose
Bit 1	Even parity over bits 2 to 13
Bits 2 to 9	Facility Code (0 to 255); Bit 2 is MSB
Bits 10 to 25	ID Number (0 to 65,535); Bit 10 is MSB
Bit 26	Odd parity over bits 14 to 25



HISTORY: GROWTH

- Magnetic swipe and Wiegand cards became widespread through the 1970s and 1980s
- Wiegand line coding became a de facto industry standard for static credentials
- Access controller panels were generally well-built for a lifetime of service, leading to high up-front prices.
 - Created a culture of “buy once, cry once”, but cemented the influence of static credentials and Wiegand-style line coding
- Upgrading beyond static credentials requires completely new hardware and different mental models from operators

HISTORY: CONTACTLESS

- As adoption grew, so did the user population and the challenges that came with it.
 - Magnetic and Wiegand cards require a human moving the card through a reader slot with consistent velocity.
- Integrated circuits were becoming increasingly prevalent, smaller, more powerful, and cheaper
- Destron/IDI developed LF originally for asset tracking and animal identification
- Hughes Aircraft was using both mag stripe readers access control and proximity tags for inventory, and saw an opportunity.
 - A new subsidiary was formed: Hughes Identification Device

HISTORY: HID

- Formed in 1991. From the early days, focused on access control.
- Access Control marketplace was ripe for disruption: multivendor parts, but rarely vertically integrated.
- Interoperability was expected in the market
- The natural MVP: the proximity card (“prox card”)
 - static credentials that can be encoded with Wiegand-style line signaling
 - Re-use the components and relationships from aircraft inventory-tracking experience
- Early investments in channel partnerships and training integrators built influence.

VULNERABILITY

- Credentials are static pieces of information, like a password.
- Credentials are wirelessly transmitted to any device that is nearby and asks for them.
 - Are sent repeatedly and quickly, without encryption.
 - No user interaction
- Static information is easily copied and cloned.
- Credentials sent between reader and control panel are unprotected
- Credentials tend to exhibit numerical adjacency and serial-like numbering

SECURITY?

- If these are so vulnerable, why do we use them?
- They “feel” secure because we lack the ability to perceive how they are operating.
 - Security Through Obscurity
- The ability to emulate and clone tags was out of reach for most people for many, many years.
- Vendors have diversified variations of the same concept, but with small tweaks to the signal

ICLASS?

- HID has deep influence and mostly rests on their laurels
- iClass was an attempt at HID in 2007 to add modern cryptography to their prox cards
 - In the name of backwards compatibility, kept static credentials, but stored encrypted on a card.
 - In the name of ease-of-deployment, “secret” static keys were installed in the firmware of their readers.
 - Static credentials are read, decrypted and sent unencrypted over the wire
- iClass Elite® Program
 - Pay HID a ton of money for a custom build of firmware for your readers with your very own unique, static key

MORAL OF THE STORY?

- Don't roll your own crypto.
- Security through obscurity is not security
- Even your temporary ideas and concepts can end up influencing your product for generations to come. Please design thoughtfully.

TODAY'S WORKSHOP

- Operate a pretend system based on static credentials on LF “prox” fobs
- I will run an app with two modes:
 - Enrollment mode: program blank fobs, using the HID 26-bit Wiegand format, saving the entries to a database
 - Reader mode: repeated poll for fobs, looking up the ID in the database, and displaying access granted/denied
- You will
 - Receive and enroll a fob in the database. Pick a name. Receive a blank fob.
 - Use the Proxmark3 instrument to clone your fob on to the other.
 - Come and test your clone and see that it works.

ATA5577C – Read/Write LF RFID IDIC 100 kHz to 150 kHz

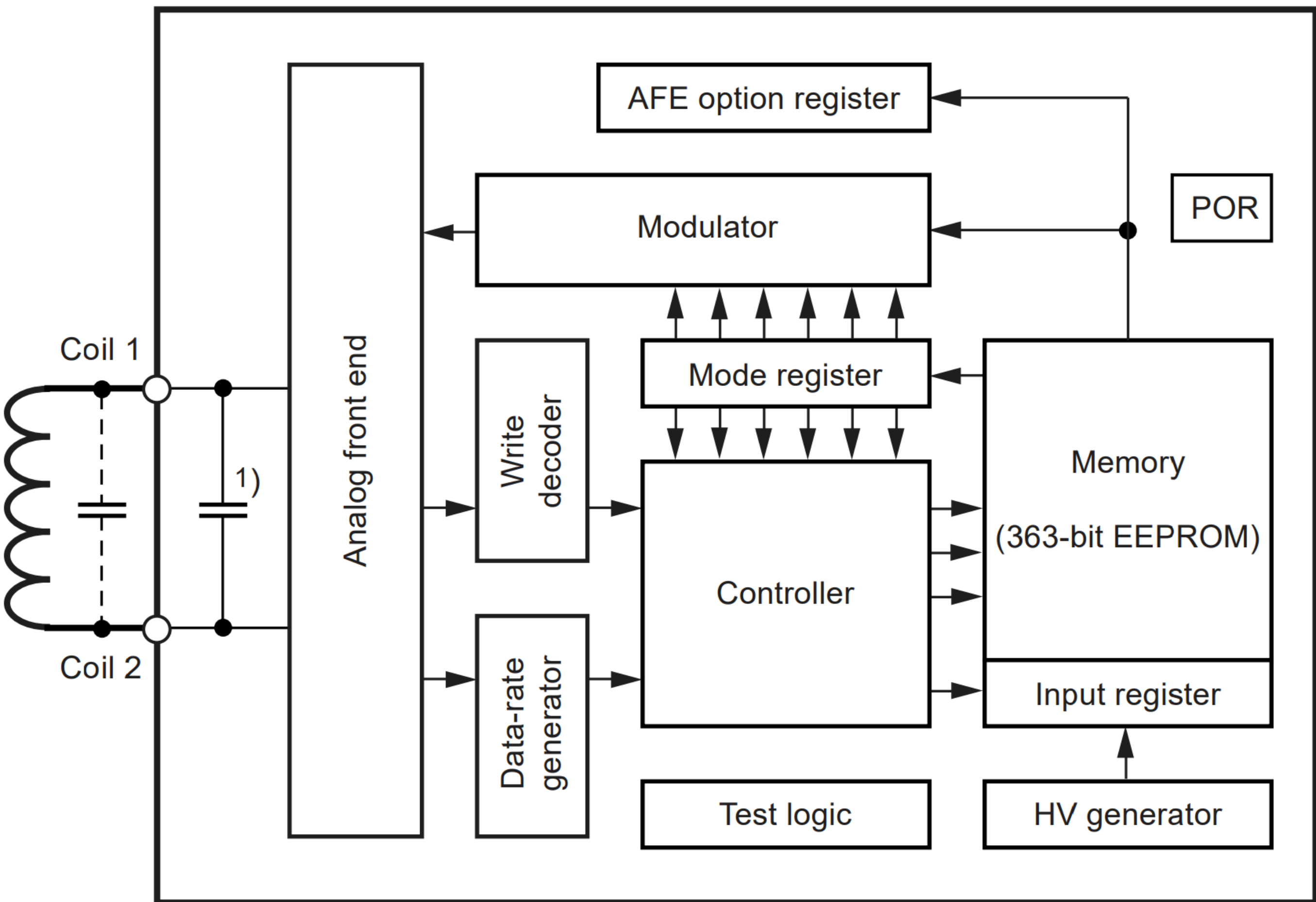
Introduction

The ATA5577C is a contactless read/write Identification Integrated Chip (IDIC[®]) for applications in the 125 kHz or 134 kHz frequency band. A single coil connected to the chip serves as the IC's power supply and bidirectional communication interface. The antenna and chip together form a transponder or tag. The on-chip 363-bit EEPROM (11 blocks with 33 bits each) is read and written blockwise from a base station (reader). Data are transmitted from the IDIC (uplink) using load modulation. This is achieved by damping the RF field with a resistive load between the two terminals, "Coil 1 and Coil 2". The IC receives and decodes serial base station commands (downlink), which are encoded as 100% amplitude modulated On-Off Keying (OOK) Pulse Interval Encoded (PIE) bit streams.

Features

These are the following features of read/write LF RFID IDIC 100 kHz to 150 kHz:

- Contactless Power Supply
- Contactless Read/Write Data Transmission
- Radio Frequency f_{RF} from 100 kHz to 150 kHz
- Basic mode or Extended mode
- Compatible with T5557, ATA5567
- Replacement for e5551/T5551 in Most Common Operation modes
- Configurable for ISO/IEC 11784/785 Compatibility
- Total 363 Bits EEPROM Memory: 11 Blocks (32 bits + 1 lock bit):
 - 7 × 32 bits EEPROM user memory, including 32-bit password memory



PROXMARK3

- Install the client CLI
 - macOS Installation Instructions
 - `brew tap RfidResearchGroup/proxmark3`
 - `brew install proxmark3`
- Plug it into your laptop and run “pm3”

[usb] pm3 --> hw version

[Proxmark3 RFID instrument]

[Client]

Iceman/master/v4.18994-suspect 2024-09-10 15:40:28 4ecb7df89
compiled with..... Clang/LLVM Apple LLVM 16.0.0 (clang-1600.0.26.3)
platform..... OSX / aarch64
Readline support..... present
QT GUI support..... present
native BT support..... absent
Python script support..... present
Lua SWIG support..... present
Python SWIG support..... present

[Proxmark3]

firmware..... PM3 GENERIC

[ARM]

bootrom: Iceman/master/v4.18341-8-g529d57a18-suspect 2024-05-01 18:04:23 20d6f7f37
os: Iceman/master/v4.18994-317-g3f5266b19-dirty-suspect 2024-10-22 21:50:41 26ade39b8
compiled with GCC 10.3.1 20210621 (release)

[FPGA]

fpga_pm3_hf.ncd image 2s30vq100 2024-02-03 15:12:20
fpga_pm3_lf.ncd image 2s30vq100 2024-02-03 15:12:10
fpga_pm3_felica.ncd image 2s30vq100 2024-02-03 15:12:41
fpga_pm3_hf_15.ncd image 2s30vq100 2024-02-03 15:12:31

[Hardware]

--= uC: AT91SAM7S512 Rev A
--= Embedded Processor: ARM7TDMI
--= Internal SRAM size: 64K bytes
--= Architecture identifier: AT91SAM7Sxx Series
--= Embedded flash memory 512K bytes (64% used)

[usb] pm3 --> ?

help	Use ` <b><command></b> help` for details of a command
prefs	{ Edit client/device preferences... }
----- Technology -----	
analyse	{ Analyse utils... }
data	{ Plot window / data buffer manipulation... }
emv	{ EMV ISO-14443 / ISO-7816... }
hf	{ High frequency commands... }
hw	{ Hardware commands... }
lf	{ Low frequency commands... }
nfc	{ NFC commands... }
piv	{ PIV commands... }
reveng	{ CRC calculations from RevEng software... }
smart	{ Smart card ISO-7816 commands... }
script	{ Scripting commands... }
trace	{ Trace manipulation... }
wiegand	{ Wiegand format manipulation... }
----- General -----	
auto	Automated detection process for unknown tags
clear	Clear screen
hints	Turn hints on / off
msleep	Add a pause in milliseconds
rem	Add a text line in log file
quit	
exit	Exit program

[usb] pm3 --> lf ?

```
help          This help
-----
awid           { AWID RFIDs... }
cotag          { COTAG CHIPS... }
destron        { FDX-A Destron RFIDs... }
em             { EM CHIPS & RFIDs... }
fdxb           { FDX-B RFIDs... }
gallagher      { GALLAGHER RFIDs... }
gproxii        { Guardall Prox II RFIDs... }
hid            { HID Prox RFIDs... }
hitag          { Hitag CHIPS... }
idteck         { Idteck RFIDs... }
indala         { Indala RFIDs... }
io             { ioProx RFIDs... }
jablotron      { Jablotron RFIDs... }
keri           { KERI RFIDs... }
motorola       { Motorola Flexpass RFIDs... }
nedap          { Nedap RFIDs... }
nexwatch       { NexWatch RFIDs... }
noralsy        { Noralsy RFIDs... }
pac            { PAC/Stanley RFIDs... }
paradox        { Paradox RFIDs... }
pcf7931        { PCF7931 CHIPS... }
presco         { Presco RFIDs... }
pyramid        { Farpointe/Pyramid RFIDs... }
securakey      { Securakey RFIDs... }
ti             { TI CHIPS... }
t55xx          { T55xx CHIPS... }
viking         { Viking RFIDs... }
visa2000       { Visa2000 RFIDs... }
-----
config         Get/Set config for LF sampling, bit/sample, decimation, frequency
cmdread        Modulate LF reader field to send command before read
read           Read LF tag
search         Read and Search for valid known tag
sim            Simulate LF tag from buffer
simask         Simulate ASK tag
-----
```

```
[usb] pm3 --> lf hid read
[+] [H10301  ] HID H10301 26-bit          FC: 15  CN: 24859  parity ( ok )
[+] [ind26    ] Indala 26-bit            FC: 246  CN: 283  parity ( ok )
[=] found 2 matching formats
[+] DemodBuffer:
[+] 1D555955556556A9A5595A69

[=] raw: 000000000000000020041ec236
[usb] pm3 --> wiegand decode --raw 000000000000000020041ec236
[+] [H10301  ] HID H10301 26-bit          FC: 15  CN: 24859  parity ( ok )
[+] [ind26    ] Indala 26-bit            FC: 246  CN: 283  parity ( ok )
[=] found 2 matching formats
[usb] pm3 --> wiegand decode --bin 00000111101100001000110110
[=] Input bin len... 26
[+] [H10301  ] HID H10301 26-bit          FC: 15  CN: 24859  parity ( ok )
[+] [ind26    ] Indala 26-bit            FC: 246  CN: 283  parity ( ok )
[=] found 2 matching formats
[usb] pm3 --> █
```

```
[usb] pm3 --> lf t55xx detect
[=] Chip type..... T55x7
[=] Modulation..... FSK2a
[=] Bit rate..... 4 - RF/50
[=] Inverted..... Yes
[=] Offset..... 34
[=] Seq. terminator... No
[=] Block0..... 00107060 (auto detect)
[=] Downlink mode..... default/fixed bit length
[=] Password set..... No
```

```
[usb] pm3 --> lf t55xx info
```

```
[=] --- T55x7 Configuration & Information -----
[=] Safer key : 0
[=] reserved : 0
[=] Data bit rate : 4 - RF/50
[=] eXtended mode : No
[=] Modulation : 7 - FSK 2a RF/10 RF/8
[=] PSK clock frequency : 0 - RF/2
[=] AOR - Answer on Request : No
[=] OTP - One Time Pad : No
[=] Max block : 3
[=] Password mode : No
[=] Sequence Terminator : No
[=] Fast Write : No
[=] Inverse data : No
[=] POR-Delay : No
[=] -----
[=] Raw Data - Page 0, block 0
[=] 00107060 - 000000000000100000111000001100000
[=] --- Fingerprint -----
[+] Config block match : HID 26b (ProxCard), Paradox, AWID
```

[usb] pm3 --> lf t55 clonehelp

For cloning specific techs on T55xx tags, see commands available in corresponding LF sub-menus, e.g.:

- lf awid clone
- lf destron clone
- lf em 410x clone
- lf fdxb clone
- lf gallagher clone
- lf gproxii clone
- lf hid clone
- lf idteck clone
- lf indala clone
- lf io clone
- lf jablotron clone
- lf keri clone
- lf motorola clone
- lf nedap clone
- lf nexwatch clone
- lf noralsy clone
- lf pac clone
- lf paradox clone
- lf presco clone
- lf pyramid clone
- lf securakey clone
- lf viking clone
- lf visa2000 clone

```
[usb] pm3 --> lf hid clone --help
```

clone a HID Prox tag to a T55x7, Q5/T5555 or EM4305/4469 tag.
Tag must be on the antenna when issuing this command.

usage:

```
lf hid clone [-h] [-w <format>] [--fc <dec>] [--cn <dec>] [-i <dec>] [-o <dec>] [-r <hex>] [--q5] [--em]
               [--bin <bin>]
```

options:

-h, --help	This help
-w, --wiegand <format>	see <code>`wiegand list`</code> for available formats
--fc <dec>	facility code
--cn <dec>	card number
-i <dec>	issue level
-o, --oem <dec>	OEM code
-r, --raw <hex>	raw bytes
--q5	optional - specify writing to Q5/T5555 tag
--em	optional - specify writing to EM4305/4469 tag
--bin <bin>	Binary string i.e 0001001001

examples/notes:

lf hid clone -r 2006ec0c86	-> write raw value for T55x7 tag (HID 10301 26 bit)
lf hid clone -r 2e0ec00c87	-> write raw value for T55x7 tag (HID Corporate 35 bit)
lf hid clone -r 01f0760643c3	-> write raw value for T55x7 tag (HID P10001 40 bit)
lf hid clone -r 01400076000c86	-> write raw value for T55x7 tag (HID Corporate 48 bit)
lf hid clone -w H10301 --fc 118 --cn 1603	-> HID 10301 26 bit, encode for T55x7 tag
lf hid clone -w H10301 --fc 118 --cn 1603 --q5	-> HID 10301 26 bit, encode for Q5/T5555 tag
lf hid clone -w H10301 --fc 118 --cn 1603 --em	-> HID 10301 26 bit, encode for EM4305/4469

```
[usb] pm3 --> lf hid read
[usb] pm3 --> lf t55 detect
[=] Chip type..... T55x7
[=] Modulation..... ASK
[=] Bit rate..... 2 - RF/32
[=] Inverted..... No
[=] Offset..... 33
[=] Seq. terminator... Yes
[=] Block0..... 000880E0 (auto detect)
[=] Downlink mode..... default/fixed bit length
[=] Password set..... No
```

```
[usb] pm3 --> lf hid clone --wiegand H10301 --fc 12 --cn 3456
```

```
[=] Preparing to clone HID tag
```

```
[+] [H10301 ] HID H10301 26-bit FC: 12 CN: 3456 parity ( ok )
```

```
[+] Done!
```

```
[?] Hint: try `lf hid reader` to verify
```

```
[usb] pm3 --> lf hid read
```

```
[+] [H10301 ] HID H10301 26-bit FC: 12 CN: 3456 parity ( ok )
```

```
[+] [ind26 ] Indala 26-bit FC: 192 CN: 3456 parity ( ok )
```

```
[=] found 2 matching formats
```

```
[+] DemodBuffer:
```

```
[+] 1D55595555655695569A5556
```

```
[=] raw: 00000000000000002004181b01
```

```
[usb] pm3 --> █
```